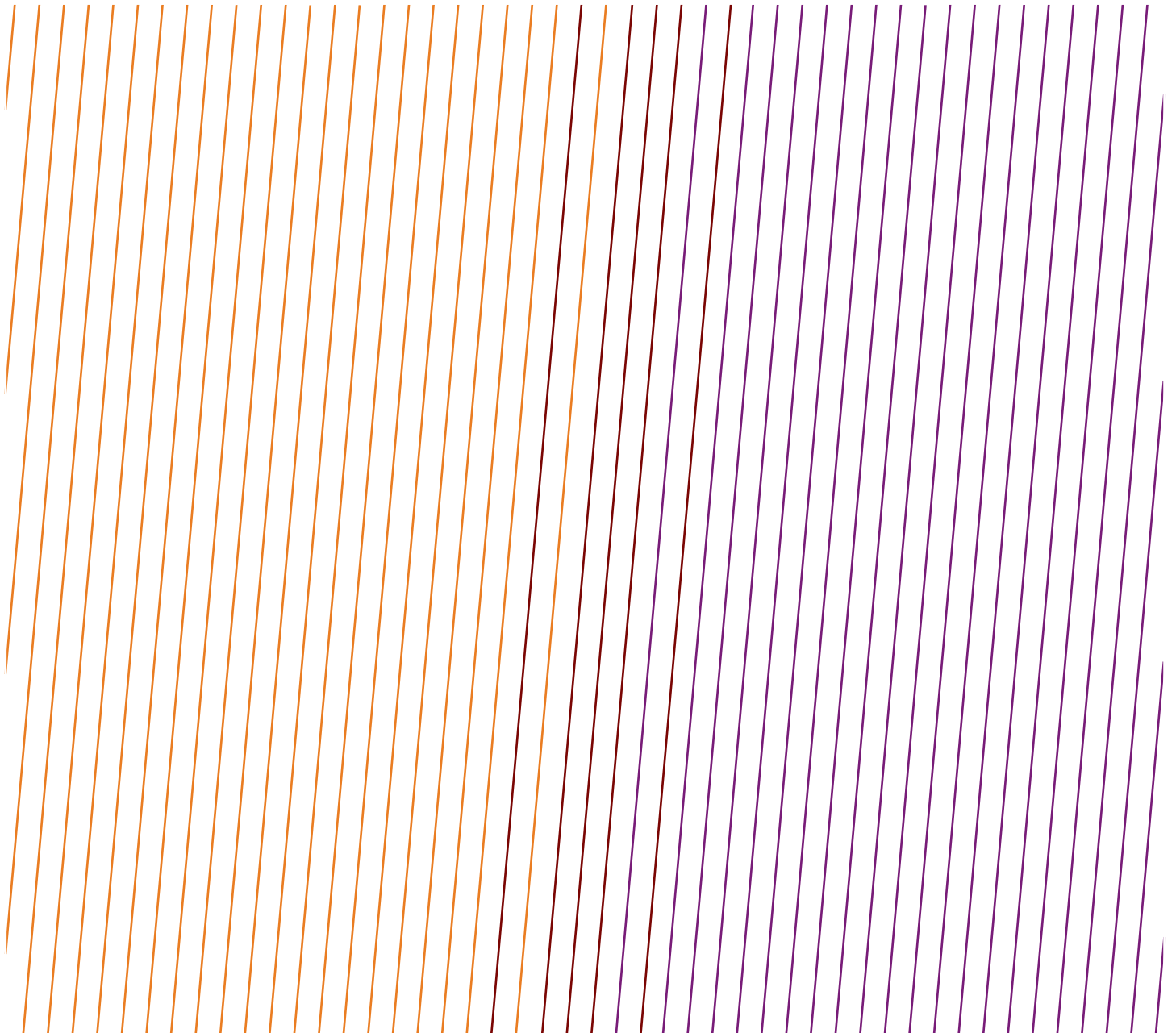


DATA SERIES

Safety performance indicators – 2024 data – Fatal incident reports



Acknowledgements

IOGP thanks those companies that have participated in the data collection programme.

This report was produced by the Safety Committee.

Feedback

IOGP welcomes feedback on our reports: publications@iogp.org

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DATA SERIES

Safety performance indicators – 2024 data
– Fatal incident reports

Revision history		
VERSION	DATE	AMENDMENTS
1.0	June 2025	First release

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AFRICA ONSHORE

DATE: 07 Mar 2024

COUNTRY: Egypt

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Falls from height

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Confined space

SECONARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Process/equipment operator, **Body part:** Head (incl. mouth, etc.), **Nature of injury:** Unspecified, **Time in service:** 10 + years

NARRATIVE:

The main operation was performing 7" casing cement job. During displacement of the cement slurry with the rig pump, part of cementing crew was carrying out cleaning operation for batch mixer 1 and 2. After cleaning batch mixer 1, the IP and one of the operators isolated batch mixer 1 and commenced cleaning operation of batch mixer 2 which has two compartments: slurry compartment, where they mix slurry, and the spacer compartment, where they mix spacer. During the activity, while IP was using the ladder to get inside in the Batch mixer 2 compartment, where the spacer was mixed, he fell down from approximately 1 m height, hitting his head. The IP was rescued by the emergency team and transported to hospital where, after 4 months unconscious, the person passed away.

WHAT WENT WRONG:

- Access to tank was not part of the authorized cleaning activity which shall be carried out without entering the tank. Despite that, IP underestimated the risks and decided to enter anyway the tank which is a confined space and requires specific PTW and mitigation measures.
- In deciding to descend into the tank, IP did not consider that the height difference between the top of the tank and the bottom presented a potential fall from height scenario and did not contemplate use of fall protection devices.
- The cleaning of the mixers was not considered in the permit to work that was issued for the cement pumping job therefore it was outside the scope of what was formally recognized/captured.
- The IP was working with another operator on the top of the tank, he decided autonomously to enter the tank to recover the hose, no Stop Work Authority was applied.
- The tank internal ladder does not allow easy access, and the presence of the internal structure (transverse bars) posed difficulties in rescue operations.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Always follow PTW, method statement and work instructions, avoiding personal initiatives.
- Carefully evaluate the risks associated with an activity before undertaking it, especially when dealing with confined spaces. Ensure a specific confined space PTW, an entry certificate and all the mitigation measures are in place before the activity. Do not proceed if a dedicated qualified attendant (watcher) is not present close to the entrance and if rescue equipment is not ready.
- Reassess the risk in case of changes in the working environment, method statement or scope of activity.
- In case of Unsafe Act/Unsafe Condition always intervene applying the Stop Work Authority.

AFRICA ONSHORE

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

DATE: 02 Jul 2024

COUNTRY: Nigeria

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Falls from height

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Working at height

SECONARY LIFE-SAVING RULE: Confined space

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Process/equipment operator, **Body part:** Respiratory system, **Nature of injury:** Major/multiple system trauma, **Time in service:** –

NARRATIVE:

After 1 hour 20 minutes of working at height via rope access within a storage tank (confined space), an Industrial Rope Access Trade Association Level 3 certified rope access technician (RAT) had trouble when trying to descend from his position in the tank roof space, 10m above the tank base.

The Level 3 RAT (IP) was transitioning from his aidings (fixed lanyards used for traversing between roof structures), to a working rope he had rigged for his vertical descent.

The IP found himself in a position below his foot-loop, unable to reach his ascender device, and unable to disconnect his remaining fixed lanyard (cow's tail).

Eventually, the IP became suspended from his harness link in a 'belly-up' position, unable to recover for about 1 hour until a rescue attempt was made by Level 3 RATs from an unrelated work scope elsewhere on the site; by which time the IP presented no vital signs. Subsequent efforts to resuscitate the IP were unsuccessful.

WHAT WENT WRONG:

IP disregarded working time restriction. IP descended before disconnecting his final fixed lanyard (cow's tail). No Working at Height rescue plan in place. Organization and rescue capabilities of RAT team were not adapted to the rescue techniques required at the time of the event. Safety Watcher (SW) without means of communication. Safety Watcher with insufficient authority.

- Communication: Ineffective RAT team communication method (hand signals), and inadequate means of communication (no team radio). Safety Watcher without means of communication (no radio), and ineffective communication of SW's authority to work party. Lack of crew-change handover on scope-critical topics between work scope owner (SIE).
- Administrative Controls: (PTW risk control measures, appointments and site risk management expectations). Agreed mitigations for hazards were not effectively implemented or formally controlled (e.g., max confined space

AFRICA ONSHORE

entry duration). Pre-job concerns over practicality of task within time constraints was not addressed (failure to stop and reassess) Appointment of critical PTW positions was not formalized, defined or well communicated (e.g., SW duties).

- Policy: Low awareness and non-compliance; inspection service provider did not consider its requirements when proposing RAT team composition Local handover procedure not followed. 'Temporary, short term' nature of work scope owner cover influenced an informal handover which did not reflect critical work scope changes.
- Work Environment: Heat and noise hazards were not effectively mitigated.
- Work Direction: The work pack (permit to work and associated risk assessment) failed to identify critical controls that should have been verified as system pre-requisite controls prior to PTW issue (e.g., presence of rescue plan, and SW duties checks). The detailed risk assessment (RAD) value is diluted by numerous, non-task related hazards which could be better managed as site standards.
- Supervision of task/crew teamwork: The specialist nature of the service provider enabled latitude for self-assessment on technical aspects, and reduced need for direct site supervision. The criticality of an avoidable work scope owner crew-change during critical work scope was not recognized when plan timing was finalized.
- Selection of Worker(s): RAT team members selected without the competence or capability to affect a rescue for the initial rig-up phase of the inspection work scope.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Working time should be respected: Adherence and strict management of working time restrictions wherever determined as a risk mitigation.
- Qualification: Individual and team composition should be adequately qualified and capable for the task at hand, in agreement with Work Permit requirements.
- For rope access work it should be carried out by specialist companies with certified staff (IRATA or equivalent) and an IRATA team with at least one IRATA 3 (highest level of certification) and no more than one third IRATA 1.
- Technical requirements: Specialist equipment and methods should be adequately maintained and aligned with technical requirements of the work scope.
- Rescue plan: Rescue plans must be realistic, practicable, and tested.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation intentional (by individual or group)

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

AFRICA OFFSHORE

DATE: 20 Mar 2024

COUNTRY: Gabon

NUMBER OF DEATHS: 6

FUNCTION: Unspecified

CAUSE: Explosion, fire or burns

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

FATALITY

Function: Drilling, **Employer:** Company, **Occupation:** Foreman, Supervisor, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Burn, **Time in service:** –

Function: Drilling, **Employer:** Contractor, **Occupation:** Foreman, Supervisor, **Body part:** Internal organs, **Nature of injury:** Unspecified, **Time in service:** –

NARRATIVE:

Ignition and explosion during WO Well Control Incident.

WHAT WENT WRONG:

Well Control Incident and Fake ATEX certification on spot lights.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Vendor control policy revision.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate/defective tools/equipment/materials/products

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

ASIA/AUSTRALASIA ONSHORE

DATE: 03 Apr 2024

COUNTRY: Indonesia

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport – Land

PRIMARY LIFE-SAVING RULE: Driving

FATALITY

Function: Production, **Employer:** Company, **Occupation:** Other, **Body part:** Internal organs, **Nature of injury:** Crushing, **Time in service:** –

NARRATIVE:

Motor Vehicle Crash (MVC) on Toll Road.

WHAT WENT WRONG:

- Driver got microsleep while driving.
- Driver didn't inspect the car before used.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Fatigue management must be implemented to prevent worker fatigue.

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Fatigue

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate supervision

ASIA/AUSTRALASIA OFFSHORE

DATE: 02 Aug 2024
COUNTRY: Malaysia
NUMBER OF DEATHS: 2
FUNCTION: Production
CAUSE: Explosion, fire or burns
ACTIVITY: Production operations
PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Engineer, Scientist, Technician, **Body part:** Respiratory system, **Nature of injury:** Burn, **Time in service:** > 1 year < 5 years

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Other, **Body part:** Respiratory system, **Nature of injury:** Burn, **Time in service:** 0 months < 3 months

NARRATIVE:

Cargo pump explosion aboard platform supply vessel while performing liquid transfer activity. Seven personnel sustained injuries. Two deceased persons (DPs) and five injured persons (IPs).

WHAT WENT WRONG:

Under investigation.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Under investigation.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

EUROPE ONSHORE

No fatal incidents reported.

EUROPE OFFSHORE

DATE: 28 Feb 2024

COUNTRY: Norway

NUMBER OF DEATHS: 1

NUMBER OF PERMANENT IMPAIRMENT INJURIES: 1

FUNCTION: Production

CAUSE: Aviation accident

ACTIVITY: Transport – Air

PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

FATALITY

Function: Production, **Employer:** Company, **Occupation:** Other, **Body part:** Respiratory system, **Nature of injury:** Major/multiple system trauma, **Time in service:** –

PERMANENT IMPAIRMENT

PI Category: Permanent reduction in skin and/or musculoskeletal function, **PI Subcategory:** –, **Body part:** Neck/torso/spine, **Nature of injury:** Major/multiple system trauma, **Time in service:** –, **Function:** Production, **Employer:** Contractor, **Occupation:** Other, **Body part:** Neck/torso/spine, **Nature of injury:** Major/multiple system trauma, **Time in service:** –

NARRATIVE:

A search and rescue helicopter crashed. The helicopter type was Sikorsky S-92 Search and Rescue (SAR), and it was on a training flight. The accident occurred in connection with training carried out at low altitude in connection with hoisting from the sea. The accident led to the death of the SAR nurse, one of a crew of a total of six people.

WHAT WENT WRONG:

The Norwegian Safety Investigation Authority (NSIA) is still investigating the accident, and no final investigation report is yet available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

An internal investigation is conducted to improve SAR helicopter safety on the Norwegian Continental Shelf with special focus on emergency response. This investigation recommends measures related to:

- Clearer procedures for when the SAR crew must unfasten the seat belt.
- Review of the personal protective equipment for SAR crew.
- Reintroduce more frequent water training (including helicopter rollovers).
- Use of the same protective equipment during water training as that used in the helicopter.
- More realistic water training within training and drills, protective equipment and coordinated action within the organisation's SAR service and emergency preparedness.

CAUSAL FACTORS:

No causal factors allocated.

MIDDLE EAST ONSHORE

DATE: 26 Apr 2024
COUNTRY: Kurdistan Region Of Iraq
NUMBER OF DEATHS: 4
FUNCTION: Production
CAUSE: Assault or violent act
ACTIVITY: Construction, commissioning, decommissioning
PRIMARY LIFE-SAVING RULE: Other issue – no applicable rule

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Unspecified, **Nature of injury:** Unspecified, **Time in service:** Unspecified

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Unspecified, **Nature of injury:** Unspecified, **Time in service:** Unspecified

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Unspecified, **Nature of injury:** Unspecified, **Time in service:** Unspecified

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Unspecified, **Nature of injury:** Unspecified, **Time in service:** Unspecified

NARRATIVE:

At 18:43hrs the repair site of the NGL storage tank was struck by an ordnance loaded drone strike launched from an unknown location. The drone strike resulted in a mass casualty event, tragically resulting in 4 contractor fatalities and 8 contractor injuries.

WHAT WENT WRONG:

A drone attack from unknown external source that we cannot control.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Although drone attacks are beyond our control, we have identified key learnings to enhance our site's resilience:

1. We have adapted our operational schedule to avoid night time operations, having observed a pattern of attacks occurring during this period.
2. We are exploring the deployment of air defence systems to strengthen our site's security and protect against potential threats.

CAUSAL FACTORS:

No causal factors allocated.

MIDDLE EAST ONSHORE

DATE: 25 Apr 2024

COUNTRY: Kuwait

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Falls from height

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Working at height

SECONARY LIFE-SAVING RULE: Bypassing safety controls

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Manual labourer, **Nature of injury:** Organ function / loss or damage, **Time in service:** –

NARRATIVE:

Contractor crew was working to spade isolated inlet to the crude tank in preparation to carryout inspection and maintenance activities for the tank. During the cold de-bolting works of the inlet, one of the Contractor personnel fell from height resulting in extensive body damage, multiple bone fractures and internal bleeding. He passed away while transferring him to hospital.

WHAT WENT WRONG:

- No work planning by the Contractor.
- Specified controls by the Company not followed by the Contractor.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- For major activities such as tank decommissioning, drainage, isolation, spading, water filling, purging, entrance, inspection and maintenance, detailed plan shall be submitted for review and approval.
- Critical tasks shall be identified in the Plan. This must be used to assign appropriate supervision.
- Effective communications is key during the execution of these critical tasks.

CAUSAL FACTORS:

PEOPLE (ACTS): Use of Protective Methods: Inadequate use of safety systems

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective guards or protective barriers

MIDDLE EAST ONSHORE

DATE: 15 Apr 2024

COUNTRY: Oman

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Water related, drowning

ACTIVITY: Transport – Land

PRIMARY LIFE-SAVING RULE: Driving

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Production, **Employer:** Company, **Occupation:** Other, **Body part:** Respiratory system, **Nature of injury:** Unspecified, **Time in service:** –

NARRATIVE:

As three staff were travelling back home from the field for rest days they encountered a flowing wadi and decided to cross it. They got trapped in the middle of the wadi. A police helicopter was mobilised to evacuate them, and during the rescue, one individual lost their life.

WHAT WENT WRONG:

Risk taking behaviour (crossing flowing wadis), compliance failings (employees did not follow Company procedures for field departure and journey management), social pressure (employees spent part of Ramadan and Eid in the field and were under pressure from family members to return home) and delayed decision making on the crew change.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. All crew members should use company's bus for crew change.
2. Amendment of crisis management and emergency response procedure to include provision for the Head of Security to contact police stations close to and within Company Blocks during adverse weather conditions in order to gather information on weather conditions and flowing wadis.
3. Review and adjust Safety Intervention Program content to reflect lessons learned from this incident.

CAUSAL FACTORS:

PEOPLE [ACTS]: Inattention/Lack of Awareness: Improper decision making or lack of judgment

MIDDLE EAST ONSHORE

DATE: 24 Mar 2024

COUNTRY: UAE

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Dropped objects

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Manual labourer, **Body part:** Neck/torso/spine, **Nature of injury:** Major/multiple system trauma, **Time in service:** > 6 months < 1 year

NARRATIVE:

During rig down of manual tong, a Floor-man (FM) positioned the tong on the rig floor and tied a manila rope to the hang-line on the tong. When the FM removed the shackle connecting the tong to hang line, the manila rope snapped. The tong counterbalance slid on its guide bars and hit the bottom mounting plate. Both guide bars released from the bottom mounting plate resulting in the counterbalance detaching from the bars falling on the FM causing a fatal injury.

WHAT WENT WRONG:

- Lack of clarity on work suspension instructions. (Inadequate Communication).
- The team was focused on completing the work. (Delivery Mindset vs Safe Execution).
- Lack of appreciation of the limits of human capabilities, the task becomes normalized and accepted by the individual. (Situational Awareness).

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Report all incomplete procedures/work instructions.
- Importance of human performance training.
- Adhere to all Work Stop communication.
- Start work only after all pre-start checks/verification are in place.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

MIDDLE EAST ONSHORE

DATE: 25 Sep 2024

COUNTRY: UAE

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Transport – Land

PRIMARY LIFE-SAVING RULE: Driving

FATALITY

Function: Drilling, **Employer:** Company, **Occupation:** Drilling/well servicing operator, **Body part:** Head (incl. mouth, etc.), **Nature of injury:** Major/multiple system trauma, **Time in service:** > 5 years < 10 years

NARRATIVE:

After dropping off the meal for the Night Shift Crews & Assistant Rig Manager, the Light Vehicle Driver along with waiter returned to the fly camp to collect more fruit. The Rig Light Vehicle Driver picked the Fruit Plate and came to the rig location, parked his car near the mess hall and delivered the fruit plate to Rig Manager's office. While leaving the rig site, the vehicle ran over a Contractor's Employee (Crane Operator) resulting in a fatal injury.

WHAT WENT WRONG:

The Driver started journey without authorization (not taking radio operator or rig manager permission/instruction). Also, the driver has a history of violations.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Obtain authorization prior commencing the travels on rig sites.
- Ensure adherence to the Site Safety Rules and Regulations including site Traffic Management Plan.
- Follow the safe driving practices and avoid the activities which may cause distractions while driving.

CAUSAL FACTORS:

No causal factors allocated.

MIDDLE EAST OFFSHORE

DATE: 19 Feb 2024

COUNTRY: Qatar

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Confined space

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Confined space

SECONDARY LIFE-SAVING RULE: Bypassing safety controls

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Maintenance, craftsman, **Body part:** Respiratory system, **Nature of injury:** Inhalation, poisoning, intoxication, asphyxiation, drowning, **Time in service:** 0 months < 3 months

NARRATIVE:

The Injured Person (IP) was found unconscious with his head and shoulders inside an exchanger channel head which was oxygen deficient due to nitrogen purging. The IP's colleague observed the scene and promptly summoned help from a nearby Safety Officer, who in turn called for emergency services. The IP was retrieved from the exchanger channel head and first aid was initiated on site. The IP was then transported to Hospital where he remained in a critical condition and passed away on 28 February 2024.

WHAT WENT WRONG:

- Sustain Barriers: The IP entered a restricted area to perform an unplanned inspection.
- Respect Hazards: The IP exposed himself to an oxygen deficient atmosphere.
- Apply Procedures: The IP failed to:
 - Obtain the required authorisation and permits and establish its required controls.
 - Use Self-Contained Breathing Apparatus (SCBA) when entering the barricaded area with an oxygen deficient atmosphere.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Adequate planning and preparation allow identification of tasks, associated hazards, and implementation of controls to achieve safe and timely execution.
- It is important to consistently follow procedures and stop work if compliance becomes challenging. Speak up and raise concerns if procedure requirements are unclear, difficult to implement, or involve unavailable tools at site.
- To access confined spaces or its exclusion zone, a PTW is required along with the implementation of all associated controls.
- A nitrogen rich environment can rapidly displace oxygen, cause unconsciousness within seconds, and potentially cause death due to asphyxiation.
- Violating Life-Saving Rules can result in fatal consequences.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

PROCESS (CONDITIONS): Organizational: Inadequate supervision

PROCESS (CONDITIONS): Organizational: Poor leadership/organizational culture

NORTH AMERICA ONSHORE

DATE: 07 Jun 2024

COUNTRY: Mexico

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

NARRATIVE:

Fire in large-scale exploitation operation.

WHAT WENT WRONG:

Lack of communication between workers.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Reinforcement of communication during start of activities meetings.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

DATE: 24 Apr 2024

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Transport – Land

PRIMARY LIFE-SAVING RULE: Driving

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Transportation operator, **Body part:** Neck/torso/spine, **Nature of injury:** Major/multiple system trauma, **Time in service:** –

NARRATIVE:

At the intersection of two county roads, an employee of a subcontractor was driving an empty tandem axle truck when the driver was involved in a motor vehicle collision with a third-party vehicle. The tandem axle truck rolled over and, as a result of the accident, the driver did not survive. The driver of the third-party vehicle was not seriously injured.

WHAT WENT WRONG:

Not available.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Scope of operational control is a key driver of proper planning of Control of Work and safe operations and should be communicated consistently.

NORTH AMERICA ONSHORE

- Investigation report in "draft" mode until police report is assessed.
- Company influence on safety culture may diminish as subcontractor layers increase.
- There are limited opportunities for Company safety culture to cascade to subcontractors.
- County roads are built for agriculture use and have not evolved to industrial use.
- Strong relationships with business partners benefit the Company.
- Business partners must drive expectations down to their subcontractors and include administrative and field verification.

CAUSAL FACTORS:

No causal factors allocated.

DATE: 05 Feb 2024

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Struck by (not dropped object)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Bypassing safety controls

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Head (incl. mouth, etc.), **Nature of injury:** Crushing, **Time in service:** –

NARRATIVE:

During re-running of tubing into a well after repair, the travelling block of the draw works struck contractor while being lowered to rig floor.

WHAT WENT WRONG:

N/A

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

N/A

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Protective Methods: Disabled or removed guards, warning systems or safety devices

NORTH AMERICA ONSHORE

DATE: 24 Jan 2024

COUNTRY: USA

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Exposure electrical

ACTIVITY: Lifting, crane, rigging, deck operations

PRIMARY LIFE-SAVING RULE: Safe mechanical lifting

SECONDARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Internal organs, **Nature of injury:** Electric shock, **Time in service:** –

NARRATIVE:

During a Flex Line Spooling operation, a telehandler contacted an overhead powerline, energizing the telehandler, and resulting in a fire. Operator exited the telehandler uninjured. After the operator exited the telehandler, a second crewmember made contact with the telehandler and was fatally injured.

WHAT WENT WRONG:

Contact by electricity.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

1. Inadequate initial emergency response: Reviewing job-site emergency action plans and workers' responsibilities in emergency response scenarios, include immediate actions for what to do if a powerline is struck (i.e., stay in vehicle, jump, shuffle, muster, etc.), reinforce initial response safety precaution to "resist rushing in!", leaders execute emergency action drill while on location.
2. 20' strap was configured in single choke position on end connection, rather than a more appropriate shorter length: Require all off-pad work with elevated equipment whose working boom or load height may exceed 17', that a spotter must visually evaluate the anticipated path of the elevated equipment to ensure that completion of the overhead powerline checklist is not required prior to commencing work. Eliminate the use of elevated equipment to transport material. Clarify "in transit" vs "working" elevated equipment.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

NORTH AMERICA OFFSHORE

DATE: 06 Apr 2024

COUNTRY: Mexico

NUMBER OF DEATHS: 3

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Production operations

PRIMARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

Function: Production, **Employer:** Contractor, **Occupation:** Unknown, **Body part:** Neck/torso/spine, **Nature of injury:** Burn, **Time in service:** –

NARRATIVE:

Fire in the processing centre.

Due to inadequate planning of works by the Command Line and failure to communicate to the operators the manual operation of the high-pressure fuel gas heaters due to high temperature, there was a sudden loss of containment and release of gas in the heater, causing a fire and explosion. The fire spread due to the storage of hazardous chemicals (toter) located on the second and third levels of the platform and the poor coordination in emergency response.

The root cause analysis to identify the causes that led to this accident and establish recommendations is under review.

WHAT WENT WRONG:

Lack of planning of activities.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Do not prioritize production over safety.

CAUSAL FACTORS:

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

RUSSIA & CENTRAL ASIA ONSHORE

DATE: 29 Jun 2024

COUNTRY: Azerbaijan

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Struck by (not dropped object)

ACTIVITY: Lifting, crane, rigging, deck operations

PRIMARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Drilling, **Employer:** Company, **Occupation:** Other, **Body part:** Head (incl. mouth, etc.), **Nature of injury:** Crushing, **Time in service:** 10 + years

NARRATIVE:

During an inspection of the pipe laydown area, the drilling foreman discovered the rigger lying motionless near the pipe stack with a head injury. After approaching and attempting to help, he realized the rigger had died. An investigation later determined that the rigger had been attempting to replace a pipe on the stack using crowbar. While moving the pipe on top stack pipe slipped, striking him on the head and causing fatal injuries.

WHAT WENT WRONG:

- Inadequate/lack of supervision.
- Inadequate work procedure.
- Inadequate control of work and risk assessment.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- During transportation and storage, the pipeline components must be suitably separated and secured to prevent rolling, shifting, sagging and vibration.
- Supports and stack heights must comply with the applicable accident prevention regulations.
- Lifting procedure must be revised and updated.
- Increasing of risk awareness and highlighting line of fire hazards during toolbox talks and training sessions.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

RUSSIA & CENTRAL ASIA OFFSHORE

DATE: 02 Sep 2024

COUNTRY: Azerbaijan

NUMBER OF DEATHS: 1

FUNCTION: Production

CAUSE: Explosion, fire or burns

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Hot work

SECONARY LIFE-SAVING RULE: Bypassing safety controls

FATALITY

Function: Production, **Employer:** Company, **Occupation:** Process/equipment operator, **Body part:** Unspecified,

Nature of injury: Major/multiple system trauma, **Time in service:** 10 + years

NARRATIVE:

An operation involving the intermittent injection of oil into the well casing for periodic cleaning the pump compressor pipes, and the filtration zone of mechanical impurities was carried out on the deck of an offshore platform. Following this operation, a fire and explosion occurred due to ingress of gas from the well into the enclosed space of the fixed cementing truck on the platform. As a result of the explosion, the well cementing truck operator sustained severe injuries and died on the spot. Investigation found out that gas ingress occurred due to violation of the valve opening sequence procedure.

WHAT WENT WRONG:

- Inadequate control of work system and risk assessment
- Inadequate design.
- Lack of supervision.
- Failure of work area preparation for fire hazardous operation.
- Inadequate hazard communication.
- Inadequate process safety management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Risk-taking behaviour can lead to serious accidents.
- The hot work permit system should be implemented, requiring gas testing to be conducted in hazardous areas.
- ATEX certified equipment to be used in hazardous areas.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Deviation unintentional (by individual or group)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Improper use/position of tools/equipment/materials/products

PEOPLE (ACTS): Use of Protective Methods: Failure to warn of hazard

PEOPLE (ACTS): Inattention/Lack of Awareness: Improper decision making or lack of judgement

PROCESS (CONDITIONS): Protective Systems: Inadequate/defective warning systems/safety devices

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate design/specification/management of change

PROCESS (CONDITIONS): Work Place Hazards: Hazardous atmosphere (explosive/toxic/asphyxiant)

RUSSIA & CENTRAL ASIA OFFSHORE

PROCESS (CONDITIONS): Organizational: Inadequate training/competence

PROCESS (CONDITIONS): Organizational: Inadequate work standards/procedures

PROCESS (CONDITIONS): Organizational: Inadequate hazard identification or risk assessment

PROCESS (CONDITIONS): Organizational: Inadequate communication

PROCESS (CONDITIONS): Organizational: Inadequate supervision

SOUTH & CENTRAL AMERICA ONSHORE

DATE: 02 Feb 2024

COUNTRY: Argentina

NUMBER OF DEATHS: 1

FUNCTION: Drilling

CAUSE: Caught in, under or between (excl. dropped objects)

ACTIVITY: Drilling, workover, well operations

PRIMARY LIFE-SAVING RULE: Line of fire

FATALITY

Function: Drilling, **Employer:** Contractor, **Occupation:** Drilling/well servicing operator, **Body part:** Unspecified,

Nature of injury: Major/multiple system trauma, **Time in service:** > 5 years < 10 years

NARRATIVE:

During a fluid evacuation operation in a container located under the screen, two workers were recovering “supernatant fluid” by using a diaphragm transfer pump. At that moment, the container truck approached the site from the front and pinned one of the workers against the container.

WHAT WENT WRONG:

- Trapping due to frontal impact of Heavy Vehicle.
- Place physical barriers with visible and clear signage, ensure the concept of “owner” of the area/facility/process in tasks where simultaneous operations with different responsible parties converge, involving movement of heavy vehicles and people who may be in the projection and contact area.
- Strengthen monitoring and control by all levels of supervision and middle management.

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

Establish a communication protocol between the operators who perform tasks in a certain area and the drivers of vehicles that can enter them (Radio use, location entry permit).

CAUSAL FACTORS:

PEOPLE (ACTS): Inattention/Lack of Awareness: Lack of attention/distracted by other concerns/stress

SOUTH & CENTRAL AMERICA OFFSHORE

DATE: 10 May 2024

COUNTRY: Brazil

NUMBER OF DEATHS: 1

FUNCTION: Construction

CAUSE: Exposure electrical

ACTIVITY: Maintenance, inspection, testing

PRIMARY LIFE-SAVING RULE: Energy isolation

SECONDARY LIFE-SAVING RULE: Work authorization

FATALITY

Function: Construction, **Employer:** Contractor, **Occupation:** Engineer, Scientist, Technician, **Body part:** Internal organs, **Nature of injury:** Electric shock, **Time in service:** > 5 years < 10 years

NARRATIVE:

During corrective maintenance to repair leaks in the air conditioning (HVAC) system on Deck 1 of the vessel, the engine room sailor was using epoxy putty with his left hand on the compressor tubing of the air conditioning system. On the second attempt to fix the persistent leak, the system was kept energized to aid in identifying leak points, allowing the team to hear the test's pressure test noises more clearly. The team believed that only the 24 Volt control system was energized. With his right hand, without gloves, the engine room sailor grabbed the back of the compressor for support, touching the electrical junction box of the equipment, which was without its protective cover and with exposed electrical terminals. The sailor was subjected to a phase-to-ground electrical voltage of approximately 493 Volts between the palm and fingers of his right hand.

The effects of this electrical shock caused burns on the palm and fingers of the right hand and cardiac fibrillation, leading to the worker's death.

WHAT WENT WRONG:

- Gaps in the corrective maintenance of the air conditioning system at the port:
 - Maintenance, conducted the week prior to the accident by a specialized company, did not have all the necessary resources to complete the service and was cleared for operation.
- Gaps in the maintenance management system:
 - There was no preventive inspection plan for the air conditioning system's electrical system.
- Inadequate technical support:
 - The onshore maintenance support was inadequate for planning the intervention and addressing the failure in the vessel's air conditioning system.
- Task planning failure:
 - The task planning did not include consultation of system documentation (electrical system diagram, compressor manual), which was also not available onboard.
 - There was no written work instruction, and no documents were issued for risk analysis regarding work on energized equipment (PT – Work Permit, JSEA – Job Safety Analysis, ICC – Isolation Control).
- Inadequate work design:
 - The installation layout of the system is inadequate for interventions (tight space, difficult access, and poor lighting), presenting ergonomic challenges for performing maintenance.

SOUTH & CENTRAL AMERICA OFFSHORE

CORRECTIVE ACTIONS AND RECOMMENDATIONS:

- Incorporate control and monitoring of maintenance tasks performed by specialized teams, ensuring that they include: detailed scope, system and/or service criticality criteria, and reservations for cases that may prevent system release.
- Establish a routine for electrical system inspection plans for facilities, covering the necessary resources for their maintenance.
- Implement a responsibility matrix in the maintenance management process.
- Ensure that the planning of interventions in electrical systems includes work instructions for the onboard team, developed in collaboration with the executing team, along with risk analysis and control documents, necessary training, and contingency plans.
- Include necessary lockout procedures in the Work Permit system.
- Implement a pre-checklist within the work permit system for high-risk activities, including necessary training.
- Establish requirements for the use of protection devices against electrical discharges in the facility's systems.
- For new projects, ensure installations and equipment are designed to facilitate maintenance interventions, considering ergonomic assessments (ergonomics of design) for those performing the tasks.
- Conduct ergonomic assessments of work for similar activities in existing facilities, particularly those involving energized equipment and hard-to-reach locations.

CAUSAL FACTORS:

PEOPLE (ACTS): Following Procedures: Improper position (in the line of fire)

PEOPLE (ACTS): Use of Tools, Equipment, Materials and Products: Servicing of energized equipment/inadequate energy isolation

PROCESS (CONDITIONS): Tools, Equipment, Materials and Products: Inadequate maintenance/inspection/testing

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